The second law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain the direction of heat transfer and the limits of energy conversion.

SUCCESS CRITERIA - I can...

1. Predict spontaneous net heat flow.
2. Explain why moving heat from cold to hot needs an additional process.
3. Distinguish energy conservation from process direction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 0.3 kg metal cools from 91 to 31°C while 0.25 kg water warms from 11 to 31°C. Use water c=4200; neglect cup/losses. Find metal c.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $Q_{hot} = Q_{cold} + W_{in}$.
Refrigerator cycle: heat amounts are positive magnitudes. Work input moves heat from cold to hot; spontaneous flow is hot to cold.

Worked reasoning: Hot and cold blocks in contact exchange energy until thermal equilibrium. Total energy can be conserved while the temperature difference and ability to drive a heat engine decrease.

Your turn: Draw before-and-after temperature states and the spontaneous heat-flow direction.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

03 Read and interpret the model

A 50°C block touches a 10°C block. Draw net heat flow and predict their trend toward equilibrium.

50°C

10°C

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Apply the physics

During one modeled refrigerator cycle, 200 J is removed from the cold compartment using 50 J of electrical work. Draw the energy flows. Find the energy released to the room and the coefficient of performance, Q_c/W. Explain why COP is not heat-engine efficiency.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A proposed machine takes heat from one reservoir and turns all of it into work in a repeating cycle. What thermodynamic limitation is missing?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 0.4 kg object moving at 6 m/s stops through friction over 0.8 m. Find energy transformed and average opposing-force magnitude.
Show the energy changes and transfers; label units.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

Two bodies exchange heat in an isolated system and reach the same temperature. Is the total energy lost? What happens to the ability to extract work from their temperature difference?

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A refrigerator removes 200 J from a cold space while receiving 80 J electrical work. Find heat delivered to the room and explain why it is larger than 200 J.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3